Data Engineering

SETUP VS CODE

What is Visual Studio Code?

💙 Visual Studio Code is the code editor that somehow starts as "just a text editor" and ends up running your entire programming life. 😄

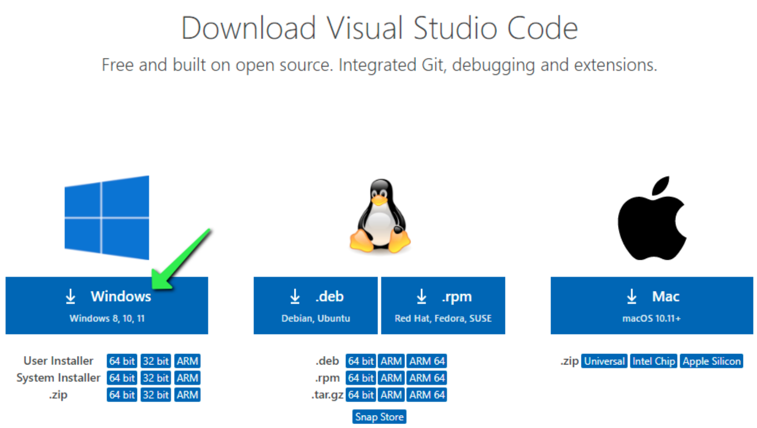
Visual Studio Code (VS Code) is a free, lightweight, and highly powerful source code editor developed by Microsoft. It is widely considered the industry standard for programmers and web developers due to its speed, built-in tools (like Git integration and debugging), and vast ecosystem of extensions.

🐧VS Code on Windows

Downloading Visual Studio Code

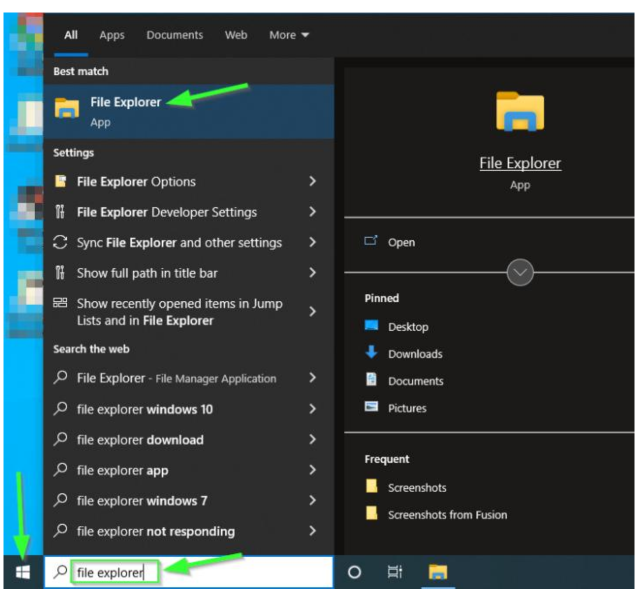
Navigate to the following link: https://code.visualstudio.com/download

Click on the blue download button that says Windows under the Windows Logo:

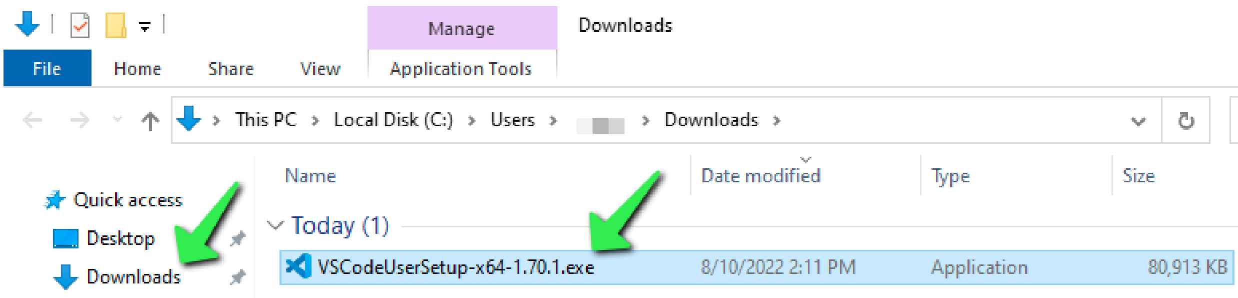


Launching the Visual Studio Code Installation Package

Once the download completes, locate it by clicking on the ⊞ Start Button and type 'File Explorer'. Launch File Explorer:

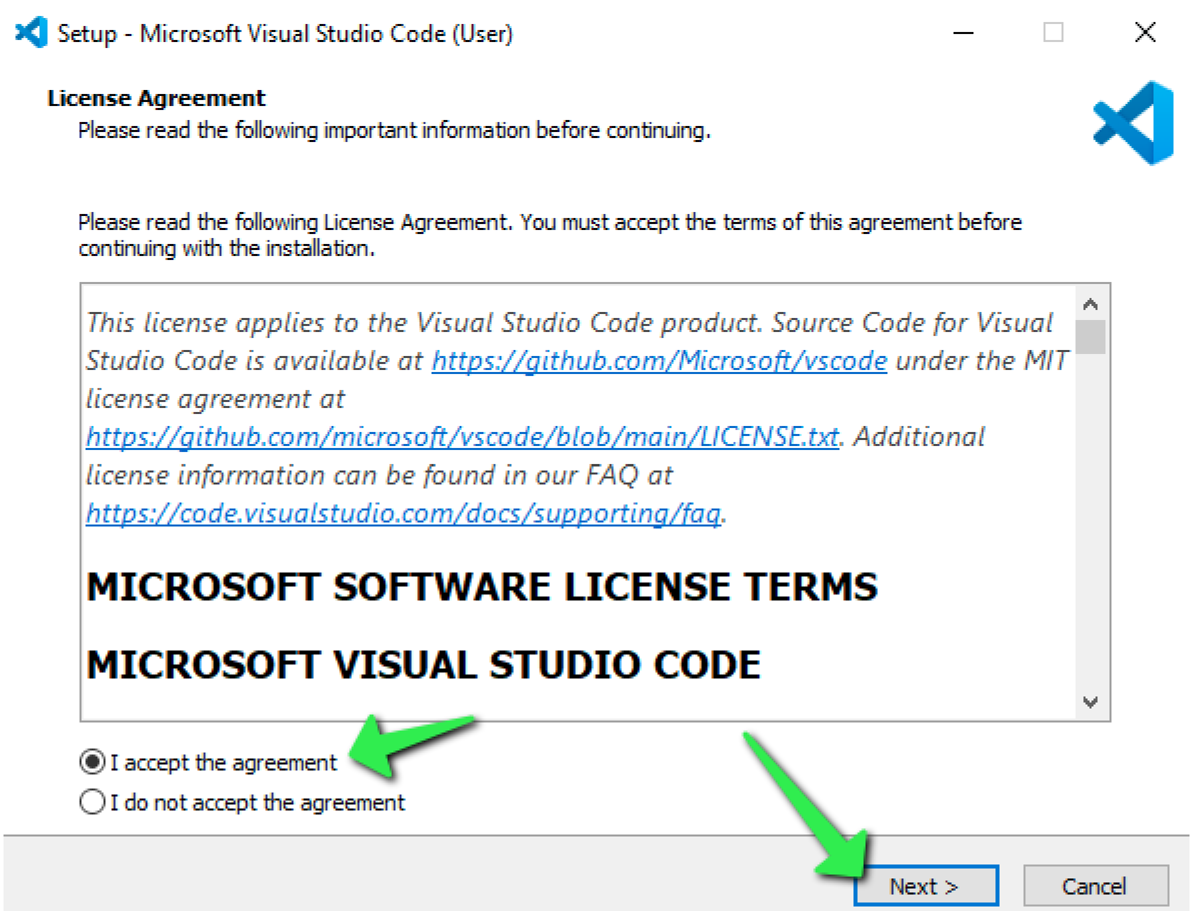


On the left side of the File Explorer window, locate and click on Downloads. Double-click on the downloaded 'VSCodeUserSetup-x64-1.x.x.exe file:

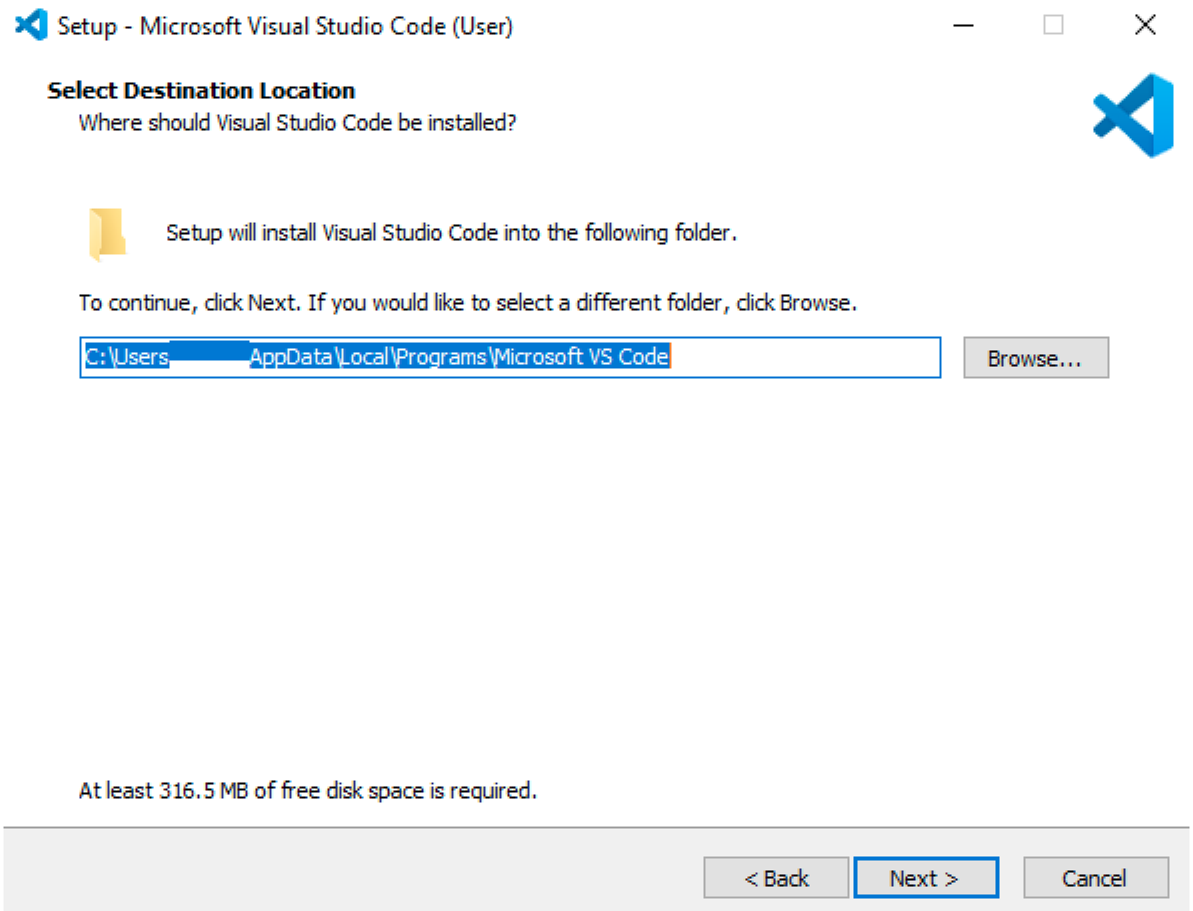


Installing

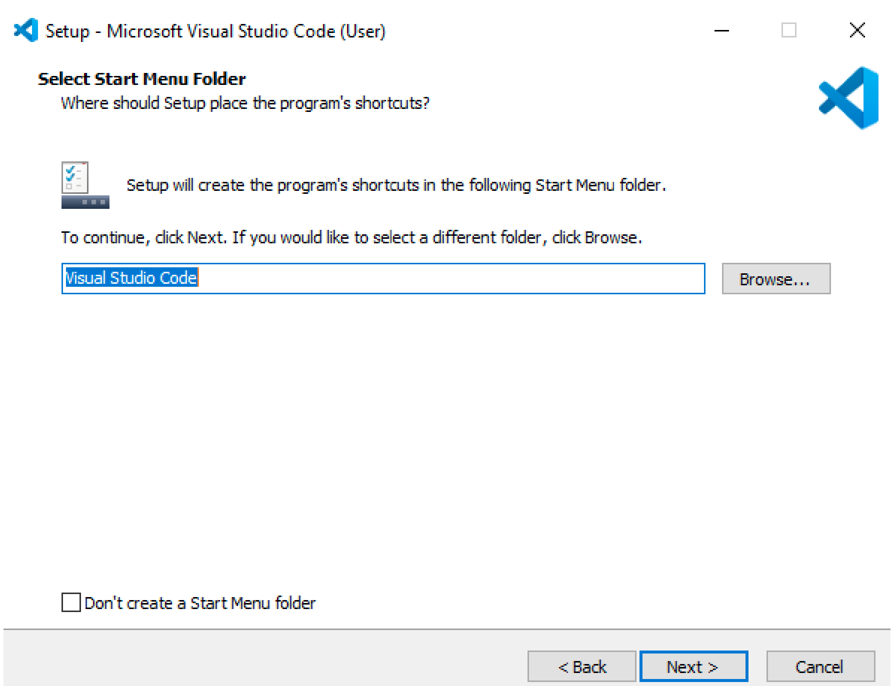
Once the installer launches, step through the installation process. First, accept the License Agreement, then click Next >.



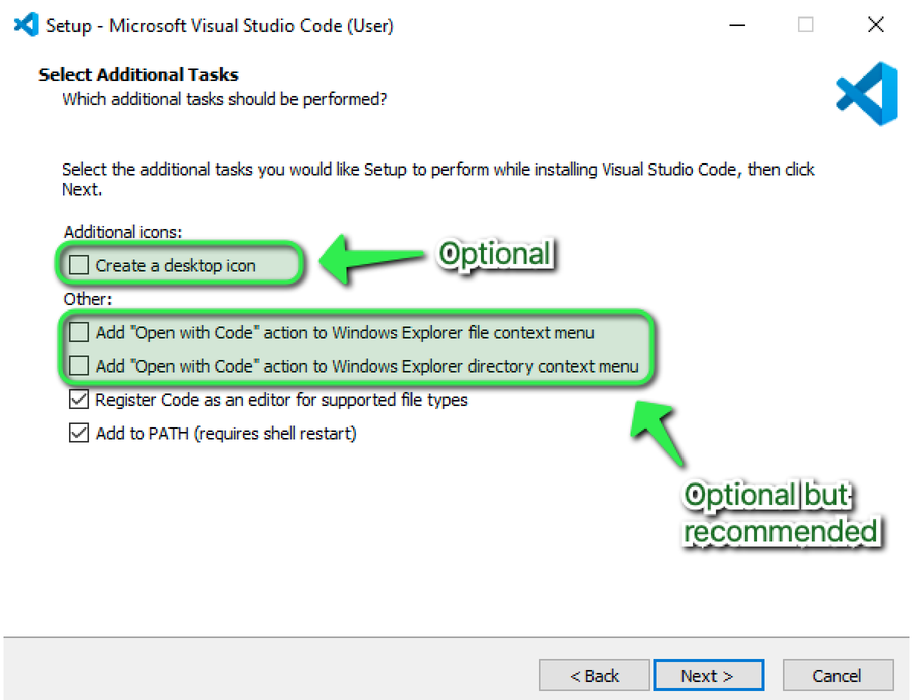
Accept the default location for installation by clicking Next >.



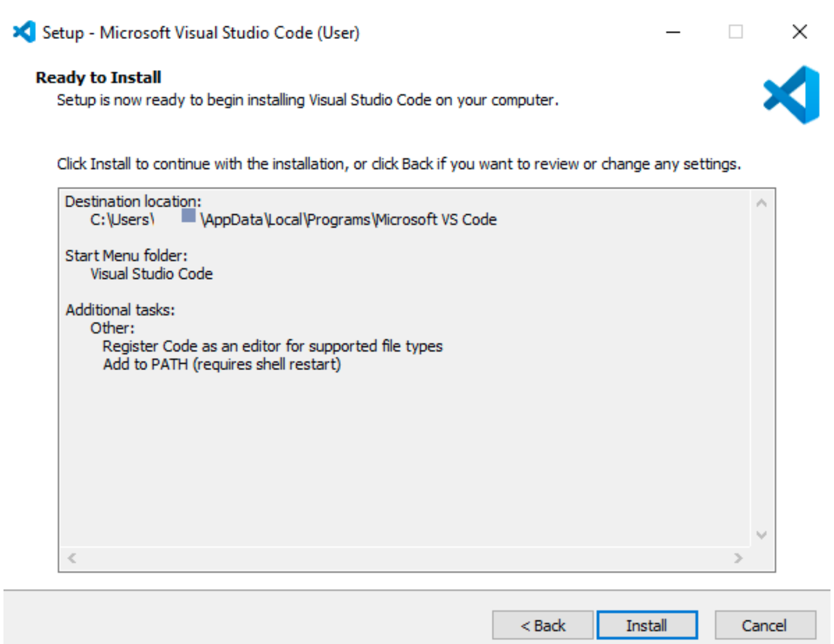
Accept the default Start Menu folder name by clicking Next >.



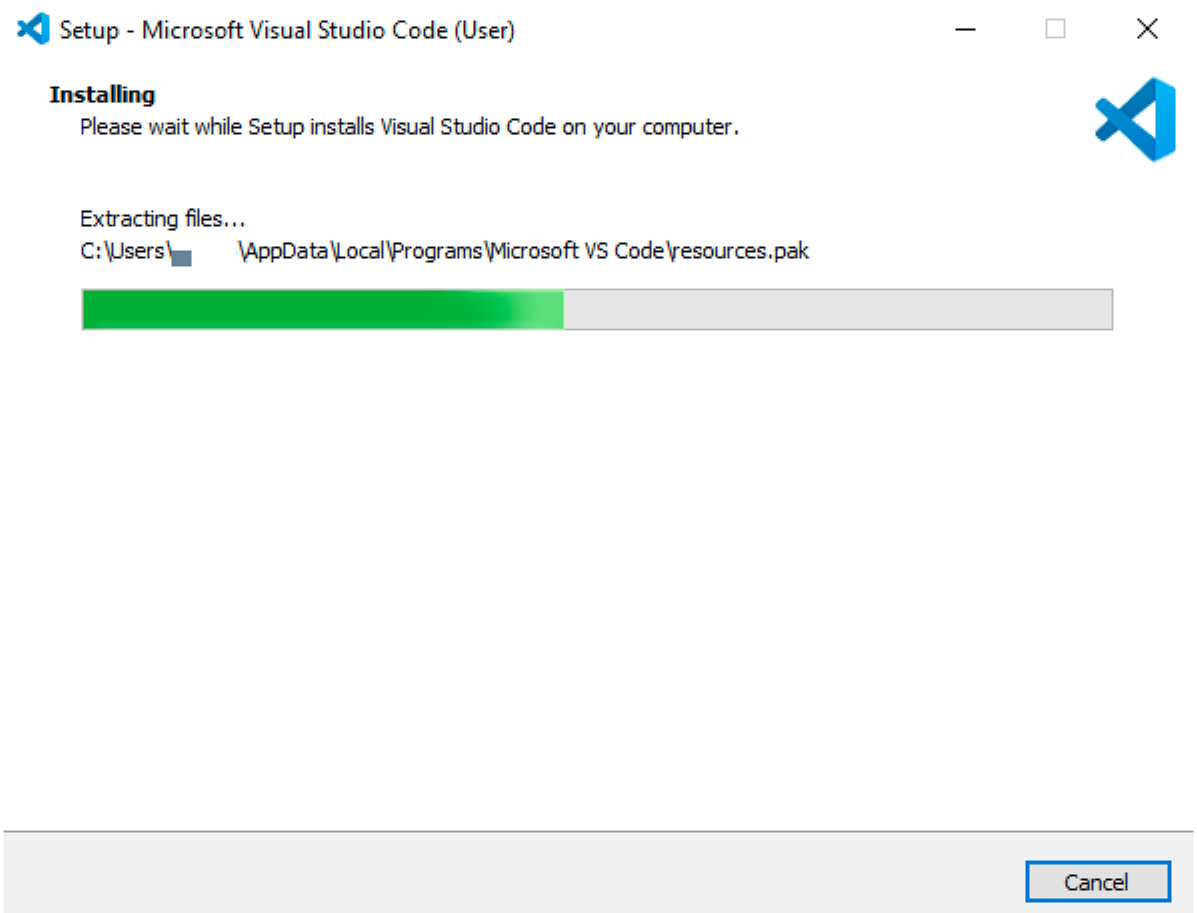
Optionally check the boxes for 'Creating a desktop icon', and adding VS Code to the Right-Click menu functionality of Windows File Explorer, then click Next >.



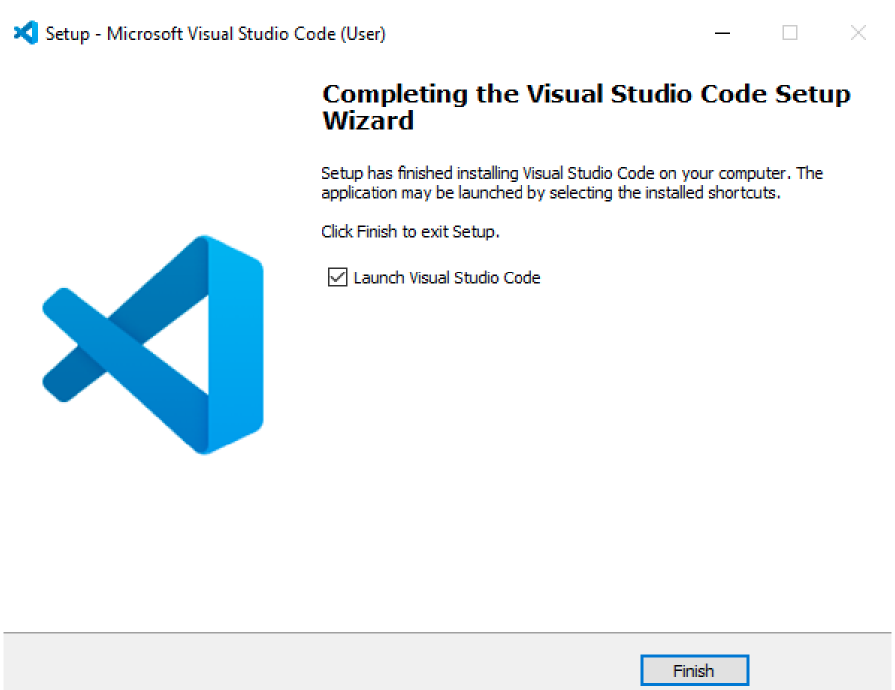
Confirm the installation options, then click Install.



The installation will proceed.

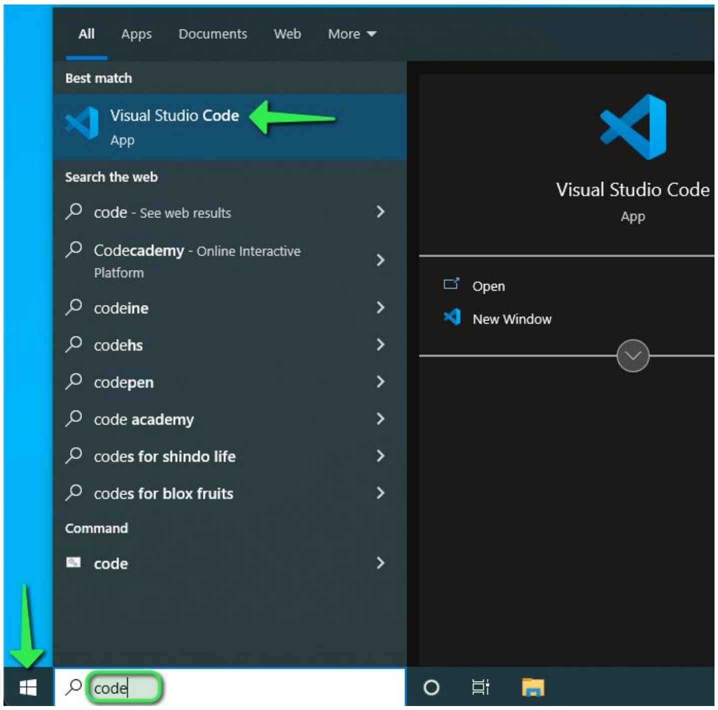


Click Finish to exit the installation and (by default) launch Visual Studio Code:

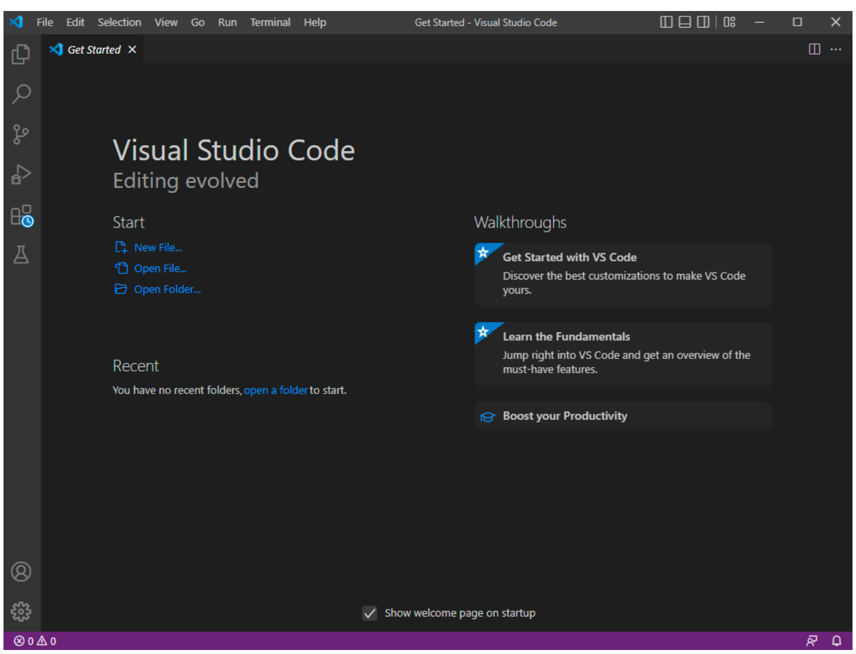


Finding VS Code after Installation

The Visual Studio Code installer will create an icon in the ⊞ Start Menu. To locate it, click on the Start Menu and search for 'Code':



Visual Studio Code will launch.

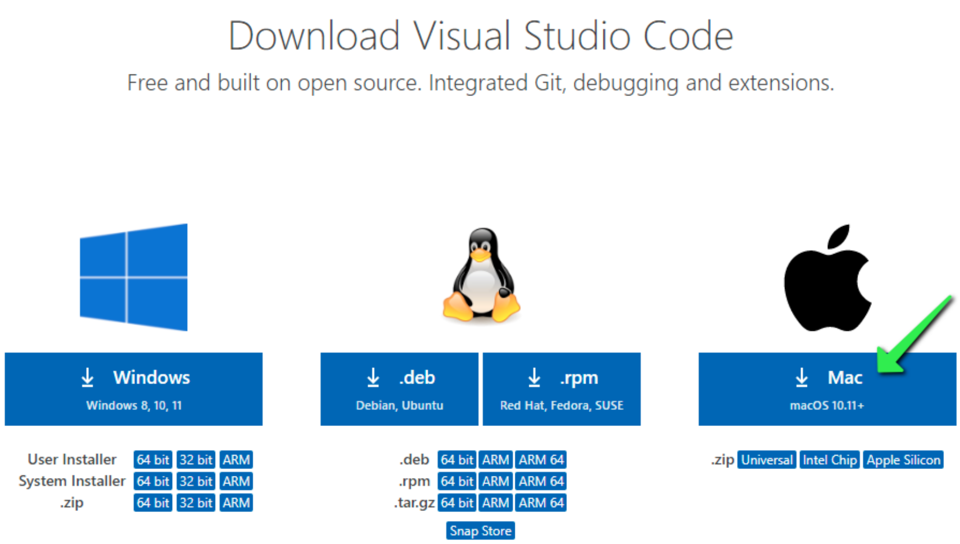


VS Code on Mac

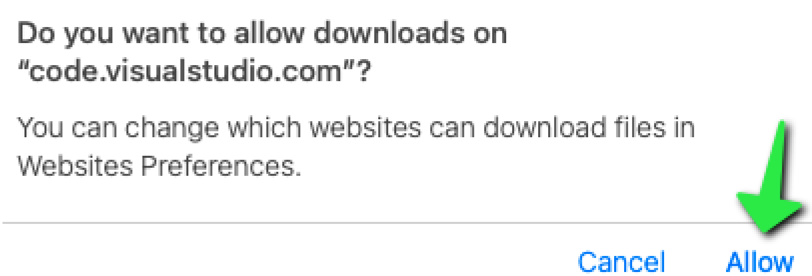
Downloading Visual Studio Code

Navigate to the following link: https://code.visualstudio.com/download

Click on the blue download button that says Mac under the Apple Logo:



Depending on the browser utilized and security settings within, there may be prompts to allow access to the Download folder or similar. Allow any prompts that are generated:

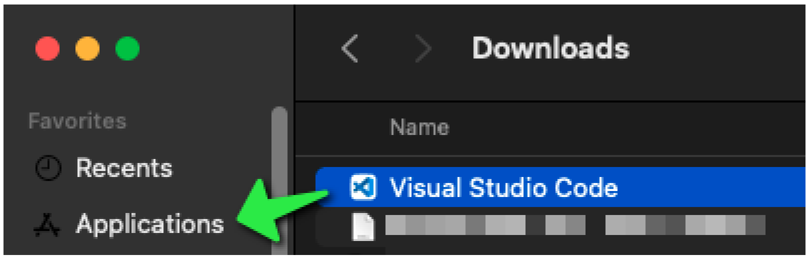


Installing Visual Studio Code

Hold down the Command ⌘ key on your keyboard and press the Space Bar and type 'Downloads' in the search field, click on the Downloads Finder icon:

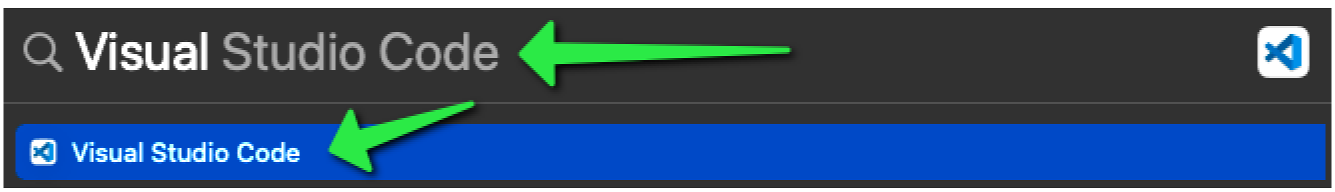
Click on Downloads on the left side of the Finder window then locate the Visual Studio Code application file. If it helps find the Visual Studio Code file, click on the Date Added column to sort by newest.

Click and drag the Visual Studio Code application to the Applications folder on the left:

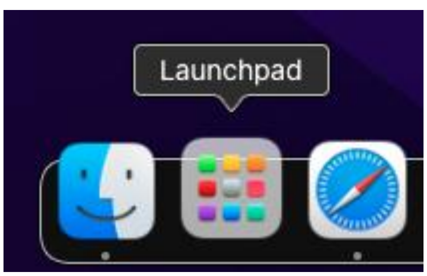


Finding VS Code after Installation

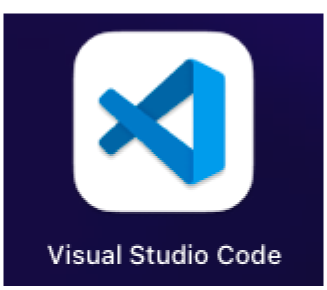
Hold down the Command ⌘ key on your keyboard and press the Space Bar and type 'Visual' in the search field, click on the Visual Studio Code icon:



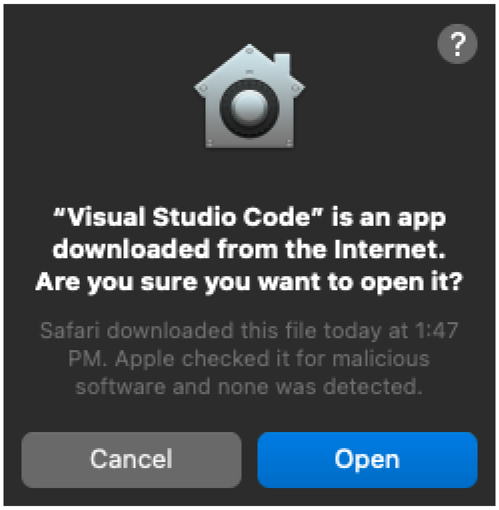
Alternatively, instead of using Spotlight Search as above, click on the Launchpad icon in the Mac OS Dock:



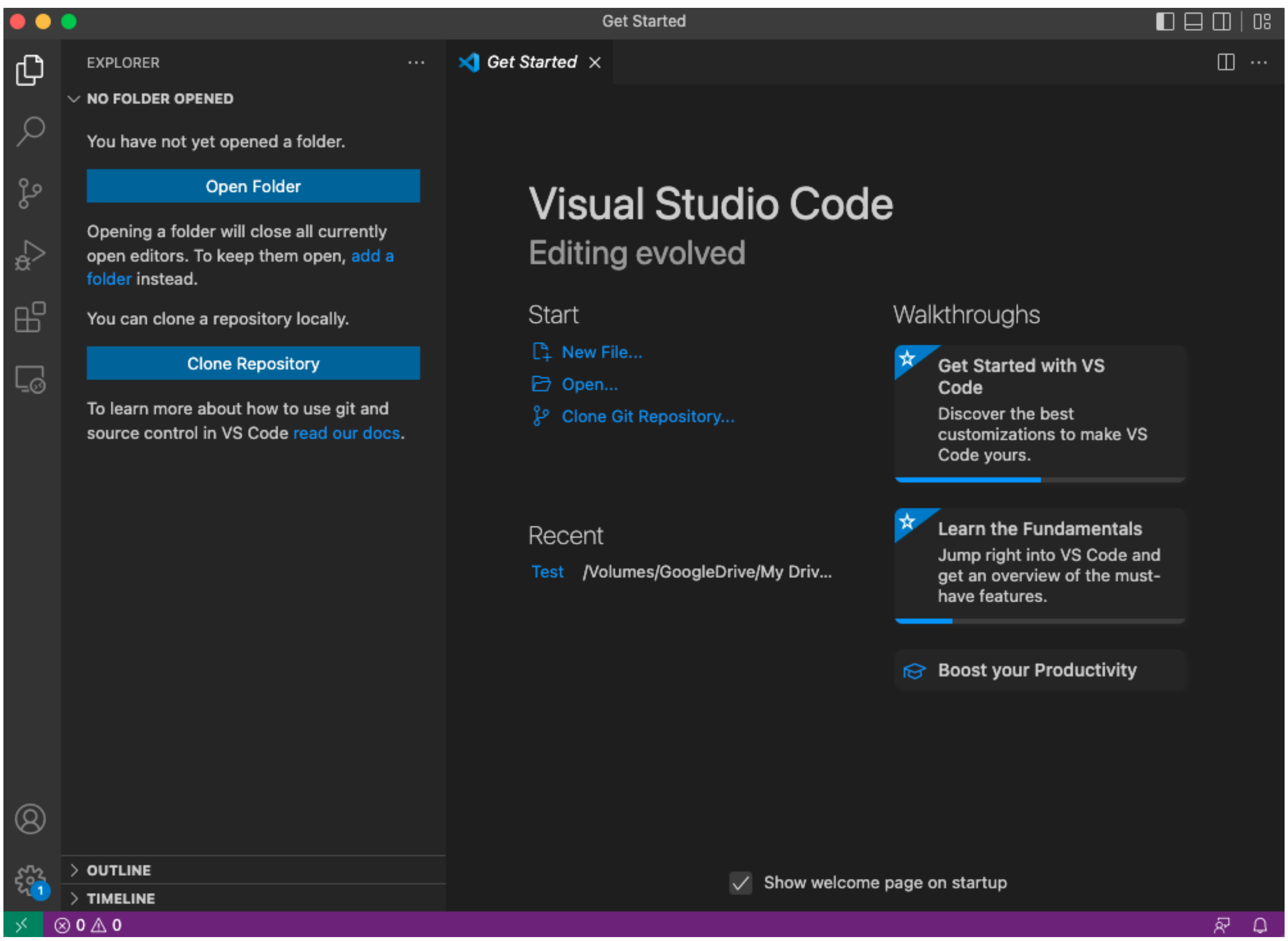
Either look for the Visual Studio Code icon, or use the Search bar at the top of the Launchpad. Click on the Visual Studio Code icon once it has been located:



Mac OS will prompt to confirm that opening a program downloaded from the internet was an intended action. Click Open:



Visual Studio Code will launch.



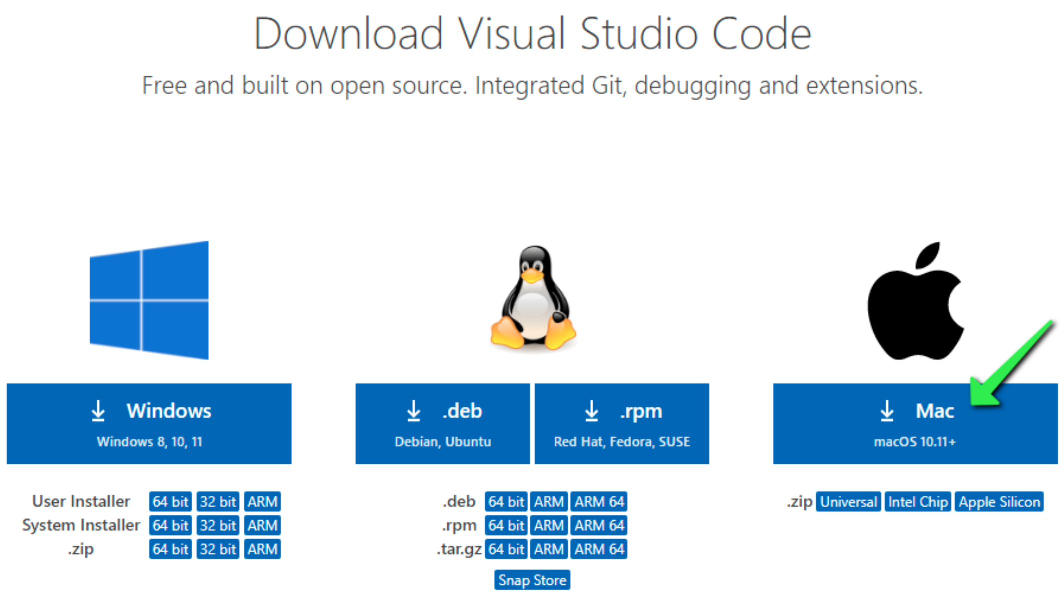
🐧 VS Code on Linux

For Debian-based Distributions

Step 1: Download and Install the .deb Package

1. Go to the [Visual Studio Code download page](#).

2. Click on the ".deb" download button.



3. Once the download is complete, open your terminal and run the following command to install the package:

```
sudo apt install ./<path-to-download>/code_1.xx.x-xxx.deb
```

Step 2: Launch Visual Studio Code

After the installation is complete, you can launch VS Code from the terminal by typing:

```
code
```

For Red Hat-based Distributions

Step 1: Download and Install the .rpm Package

1. Visit the [Visual Studio Code download page](#).
2. Click on the ".rpm" download button.

3. Once the download is complete, open your terminal and run the following command to install the package:

```
sudo rpm -i ./<path-to-download>/code-1.xx.x-xxx.rpm
```

Step 2: Launch Visual Studio Code

After the installation is complete, you can launch VS Code from the terminal by typing:

```
code
```

Customizing Visual Studio Code

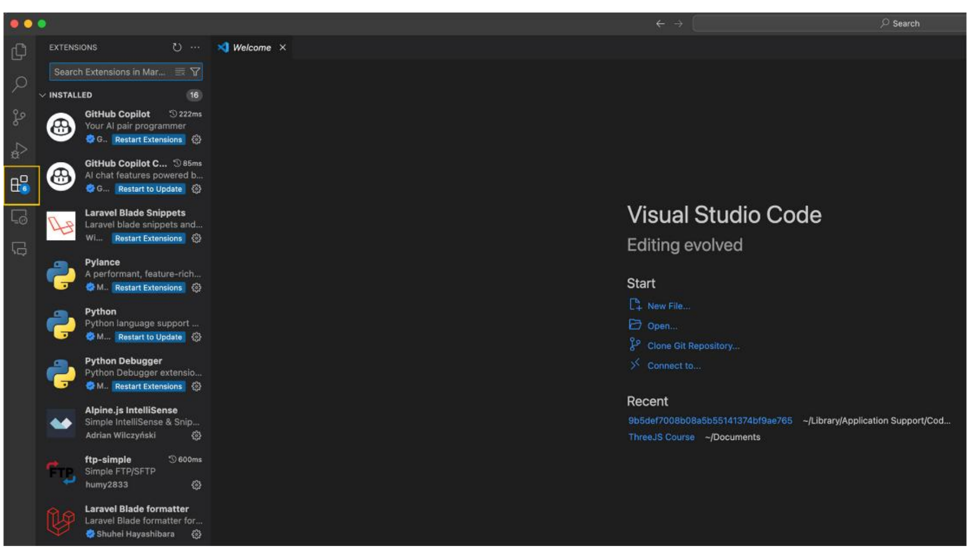
VS Code offers a wide range of customization options, providing you with the flexibility to adjust your coding environment to your personal preferences, thus creating a workspace that can increase your productivity and enhance your coding skills.

Extensions

Visual Studio Code, or VS Code, supports a plethora of extensions to enhance its functionality, allowing developers to tailor the editor according to their preferences or project requirements. Adding an extension is a simple process:

1. Click on the Extensions icon, represented by a square icon located in the Activity Bar on the side of the window. This brings up the Extensions view.

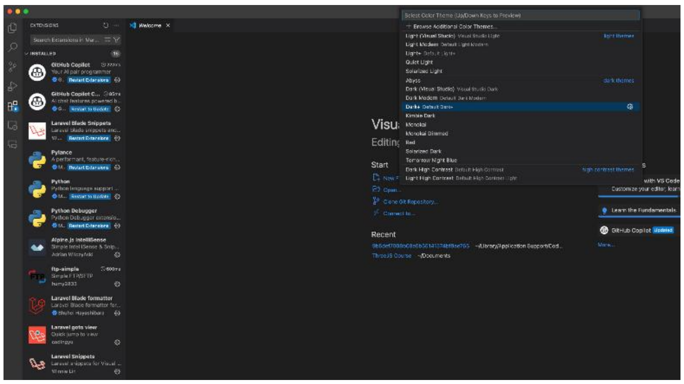
2. Once here, you can search for the extension you want to install in the search bar. When you've found the desired extension, simply click "Install" to add it to your VS Code editor.



Themes

VS Code's interface is also highly customizable, with the option to change the theme of the editor. Here's how you can do it:

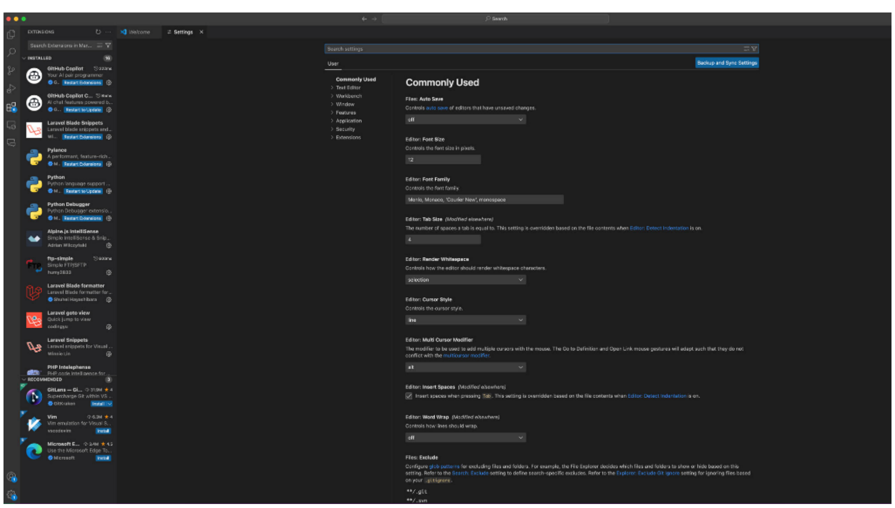
1. Navigate to File > Preferences > Color Theme. This will display a dropdown list of available themes.
2. Scroll through the list and select a theme that suits your preferences. The change will be applied immediately.



Settings

Apart from extensions and themes, you can also customize a wide array of settings in VS Code to make it truly your own:

1. Go to File > Preferences > Settings. This will open a new tab with a search bar at the top.
2. Here, you can either search for the specific setting you want to change, or browse through the categories to explore all the customization options available.



🛠️ Essential VS Code Extensions for Data Engineers

Extension	Purpose
🐍 Python	Python development, debugging, IntelliSense, and Jupyter support.
📓 Jupyter	Run and edit Jupyter notebooks directly inside VS Code.
❄️ Snowflake	Connect to Snowflake, run SQL queries, and manage databases.
🐘 PostgreSQL	Connect to PostgreSQL databases and execute SQL queries.
🗄️ SQLTools	Universal SQL client supporting PostgreSQL, MySQL, SQL Server, Oracle, SQLite, and more.
🔌 SQLTools Drivers	Database drivers used by SQLTools for various database engines.
📊 dbt Power User	Build, test, document, and visualize dbt projects.
📦 dbt Formatter	Automatically formats dbt SQL models for better readability.
🌬️ Apache Airflow	DAG syntax highlighting, templates, and Airflow development support.
🐳 Docker	Build, manage, and monitor Docker containers and images.
☁️ Remote - Containers (Dev Containers)	Develop inside Docker containers with consistent environments.
🧩 YAML	YAML syntax support, validation, and auto-completion.
📝 JSON	JSON validation, formatting, and IntelliSense.
📄 XML	XML editing, validation, and formatting.
🌐 REST Client	Send HTTP requests directly from VS Code without Postman.

Extension	Purpose
🔥 Thunder Client	Lightweight API testing similar to Postman.
🔄 GitLens	Powerful Git history, blame, and repository insights.
🌳 Git Graph	Visualize Git branches and commit history.
🎨 Prettier	Automatically format code consistently.
🖌️ Code Spell Checker	Detect spelling mistakes in code and documentation.
📦 Markdown All in One	Improve Markdown editing and documentation.
📚 Markdown Preview Enhanced	Advanced Markdown preview with diagrams and math support.
🧠 IntelliCode	AI-assisted code completion from Microsoft.
🔎 Error Lens	Displays errors and warnings inline while coding.
🎯 Better Comments	Highlight TODOs, notes, warnings, and important comments.
📂 Path Intellisense	Auto-completes filenames and directory paths.
⚙️ EditorConfig	Maintains consistent coding styles across projects.
📦 DotENV	Syntax highlighting for .env configuration files.
📜 Log File Highlighter	Makes application and Airflow logs easier to read.
📊 Excel Viewer	Open Excel files directly inside VS Code.
📄 CSV to Table	View CSV files in a clean table format.
📊 Rainbow CSV	Colorizes CSV columns and supports SQL-like queries.
🔍 Peacock	Color-code different VS Code workspaces.
📁 Project Manager	Quickly switch between multiple projects.

Extension	Purpose
⚡ Live Share	Collaborate with others in real time.
🛡️ SonarLint	Detect bugs, vulnerabilities, and code smells as you type.
🧪 Test Explorer UI	Run and manage Python tests visually.
🏗️ Makefile Tools	Build automation using Makefiles.
🐧 Remote - SSH	Connect and develop on remote Linux servers.
☁️ Azure Account	Authenticate with Azure services.
☁️ AWS Toolkit	Develop and manage AWS resources from VS Code.
☁️ Google Cloud Code	Build and deploy applications to Google Cloud.

🗃️ Database Extensions

Database	Emoji
🐘 PostgreSQL	🐘
❄️ Snowflake	❄️
🐬 MySQL	🐬
🏛️ Oracle Database	🏺
🪟 SQL Server	🪟
🍃 MongoDB	🍃
🔥 Redis	🚀
🗂️ SQLite	💾

📊 Data Engineering Stack

Tool	Emoji
🐍 Python	🐍
📓 Jupyter Notebook	🧮
🐘 PostgreSQL	🐘
❄️ Snowflake	❄️
📊 dbt	🚀
🌬️ Apache Airflow	🌪️
🐳 Docker	🐳
🌳 Git	🌲
☁️ Cloud Platforms	☁️
🌐 REST APIs	📡
📁 CSV/JSON/XML	📄

⭐ Top 15 Must-Have Extensions for Every Data Engineer

1. 🐍 Python
2. 🧮 Jupyter
3. 🗄️ SQLTools
4. 🐘 PostgreSQL
5. ❄️ Snowflake
6. 🚀 dbt Power User
7. 🌪️ Apache Airflow
8. 🐳 Docker
9. ⚡ Thunder Client

10. 🔄 GitLens
11. 🌲 Git Graph
12. 🌈 Rainbow CSV
13. 🚨 Error Lens
14. 📁 Path Intellisense
15. ✨ Prettier

These extensions provide a complete toolkit for modern data engineering workflows, covering **Python development, SQL, databases, dbt, Airflow, Docker, Git, APIs, cloud platforms, documentation, testing, and data file management**.